

| Tokenizer | 方向数 | PE | Look Bias | Top-1         | $\Delta$ Top-1  | Top-5         | $\Delta$ Top-5  | 参数量 (M) |
|-----------|-----|----|-----------|---------------|-----------------|---------------|-----------------|---------|
| Standard  | —   | ✓  | —         | 51.52%        | —               | 79.12%        | —               | 5.544   |
| Hex       | —   | ✓  | —         | 51.30%        | −0.22 pp        | 79.14%        | +0.02 pp        | 5.597   |
| Hex       | 4   | ✓  | —         | 53.56%        | +2.04 pp        | 80.40%        | +1.28 pp        | 5.486   |
| Hex       | 4   | —  | ✓         | 50.26%        | −1.26 pp        | 77.72%        | −1.40 pp        | 5.463   |
| Hex       | 4   | ✓  | ✓         | 53.70%        | +2.18 pp        | 80.46%        | +1.34 pp        | 5.501   |
| Hex       | 6   | ✓  | —         | 54.54%        | +3.02 pp        | 80.46%        | +1.34 pp        | 5.464   |
| Hex       | 6   | —  | ✓         | 52.42%        | +0.90 pp        | 79.48%        | +0.36 pp        | 5.453   |
| Hex       | 6   | ✓  | ✓         | <b>55.04%</b> | <b>+3.52 pp</b> | <b>81.80%</b> | <b>+2.68 pp</b> | 5.491   |

统一设置为  $224 \times 224$  输入、100 类、20 个训练轮次和  $2 \times \text{T4}$ ； $\Delta$  表示相对 Standard 的百分点变化；参数量为可训练参数，以百万（M）计。